Agentic Graph RAG for Explainable Multimodal Clinical Decision Support with Verifiable Evidence and Client-Side Radiographic Saliency

Dr. Madhuri Pagale, Vikram Kadam, Keshav Mittal, Sandesh Phad, and Tejas Shidam
Emails: {madhuripagale, vikramkadam23, keshavmittal23, sandeshphad23, tejasshidam23}@gmail.com

***Abstract*—While Large Language Models (LLMs) encode extensive biomedical knowledge, clinical deployment remains constrained by ungrounded hallucinations, absent literature provenance, opaque reasoning chains, uncalibrated confidence heuristics, and patient privacy risks during cloud-based multimodal inference. This paper presents MEDGUIDE AI, an explainable, multi-agent Clinical Decision Support System (CDSS) that unifies Retrieval-Augmented Generation (RAG) with structured clinical knowledge graphs, deterministic pharmacovigilance, tenant-isolated medical asset storage, and privacy-preserving client-side radiographic saliency. The proposed architecture coordinates 12 specialized workflow agents across diagnostic planning, multi-source biomedical literature retrieval (PubMed, Europe PMC, WHO, ClinicalTrials.gov), knowledge graph ontology traversal ($G = (V, E)$), deterministic pharmacology and allergy verification (openFDA, RxNorm), post-hoc safety auditing, and continuous benchmark evaluation. Diagnostic hypotheses are systematically grounded in retrieved medical sources classified under the Oxford Centre for Evidence-Based Medicine (CEBM) 5-tier schema. In addition, an in-browser DenseNet-121 neural model executing via WebAssembly generates occlusion sensitivity heatmaps on raw chest radiographs without transmitting image pixels to external servers for saliency localization, complemented by a tenant-isolated Cloudinary medical storage boundary for imaging persistence. The platform incorporates a 7-section hospital consultation document engine featuring interactive citation badges, scannable QR verification provenance, and 1-click standardized Electronic Health Record (EHR) export. The framework evaluates a multi-signal composite confidence metric categorized into ordinal bands (High, Medium, Low, Insufficient Evidence). We established an empirical evaluation protocol across 50 multi-system clinical vignettes. Experimental results indicate that Agentic Graph RAG achieves a 90.3% ± 3.94 evidence grounding rate (95% CI: [89.2%, 91.4%]), 92.2% ± 4.15 safety compliance (95% CI: [91.1%, 93.4%]), and 95.2% ± 2.76 system-defined explainability score (95% CI: [94.5%, 96.0%]), with 0.0% ghost citations ($p < 10^{-15}$), outperforming zero-shot LLMs (19.8%, 34.6%, 14.3%, 78.4% ghost citations) and standard vector RAG (64.8%, 61.3%, 56.4%, 21.2% ghost citations). The system maintains non-autonomous operational boundaries, incorporates a formal Model Facts specification, and aligns with principles outlined in the World Health Organization (WHO) 2024 guidance on Large Multi-Modal Models in healthcare.**

Dr. M. Pagale is an Assistant Professor and Research Guide in the Department of Computer Science & Engineering (AI & ML), Pimpri Chinchwad College of Engineering, Pune 411044, India (e-mail: madhuripagale@gmail.com, madhuri.pagale@pccoepune.org).

V. Kadam, K. Mittal, S. Phad, and T. Shidam are undergraduate researchers in the Department of Computer Science & Engineering (Artificial Intelligence & Machine Learning), Pimpri Chinchwad College of Engineering (PCCOE), Pune 411044, India (e-mail: {vikramkadam23, keshavmittal23, sandeshphad23, tejasshidam23}@gmail.com; {vikram.kadam23, keshav.mittal23, sandesh.phad23, tejas.shidam23}@pccoepune.org).

***Index Terms*—Clinical Decision Support Systems, Explainable AI, Retrieval-Augmented Generation, Medical Knowledge Graphs, Multi-Agent Systems, Radiographic Saliency, Edge AI, Biomedical Informatics.**

I. Introduction

A. Clinical Decision Support Background

CLINICAL Decision Support Systems (CDSS) represent an established area of biomedical informatics research [1], [2]. In modern healthcare settings, physicians encounter high patient volumes, compressed consultation windows, and rapid literature expansion, with over two million scientific articles indexed annually in MEDLINE [2], [3]. Diagnostic discrepancies and therapeutic errors continue to affect inpatient admissions, with studies reporting diagnostic error rates between 10% and 15% in complex acute presentations [2], [4]. These errors often arise from cognitive heuristics, premature diagnostic closure, atypical symptom patterns, or unflagged drug-drug and drug-allergy interactions.

Early rule-based CDSS such as MYCIN [1] and DXplain established the feasibility of encoding expert heuristics for infectious disease management and differential diagnosis. Subsequent systems embedded statistical rule triggers within Electronic Health Record (EHR) workflows [4], [5]. However, static rule-based systems exhibited limited adaptability across diverse clinical narratives and suffered from high manual maintenance overhead. Furthermore, non-selective alerting thresholds frequently induced clinical alert fatigue, resulting in override rates exceeding 90% in hospital deployments [2], [5].

B. Limitations of Generative LLMs in Medical Practice

Recent progress in biomedical Large Language Models (LLMs), including Med-PaLM 2 [6], Med-Gemini [7], and AMIE [8], has demonstrated high factual performance on medical licensing benchmarks [9]. In a double-blind crossover trial published in *Nature*, Tu et al. [8] observed that the conversational diagnostic model AMIE achieved diagnostic accuracy and consultation quality comparable to primary-care physicians across simulated clinical consultations.

Despite these capabilities, ungrounded generative models exhibit specific technical limitations when applied to direct clinical workflows:

1) **Stochastic Hallucinations:** Parametric language models sample text probabilistically, occasionally generating

fictitious clinical findings, fabricated trial identifiers (NCT numbers), or invalid PubMed IDs with fluent syntax [6], [10].

2) **Lack of Verifiable Grounding:** Monolithic LLM generations lack direct, auditable links to authoritative clinical practice guidelines (e.g., NICE, WHO, ESC, ACC/AHA) or peer-reviewed primary literature, preventing clinicians from inspecting the evidentiary basis of generated assertions [11], [12].
3) **Uncalibrated Scalar Predictions:** Standard models generate linguistic statements that may mask underlying evidence sparsity, missing clinical parameters, or contradictory findings, potentially inducing automation bias [13].
4) **Multimodal Data Privacy Concerns:** Analyzing thoracic radiographs typically involves uploading raw pixel arrays to cloud-hosted APIs. In clinical practice, this introduces data governance and patient confidentiality considerations under HIPAA and GDPR frameworks [5], [14].

These operational risks are highlighted in the *World Health Organization (WHO) 2024 Guidance on Large Multi-Modal Models in Health* [15], which recommends rigorous governance, verifiable provenance, privacy preservation, and human-in-the-loop oversight for clinical AI deployments.

C. Motivation: Transition from Parametric LLMs to Agentic Graph RAG

Retrieval-Augmented Generation (RAG) [16] connects language model generation with external, non-parametric document collections. In a baseline vector RAG pipeline, clinical text queries are embedded into dense vector spaces, and top-k document passages are retrieved via cosine similarity to provide context for generation.

Although vector RAG reduces surface hallucinations, standard dense retrieval presents limitations in complex medical reasoning:

- **Loss of Relational Structure:** Diagnostic reasoning follows relational pathways across clinical concepts:

$$\text{Disease} \xrightarrow{\text{presents}} \text{Symptom} \xrightarrow{\text{contraindicates}} \text{Drug}$$

  Unstructured passage retrieval often fails to preserve explicit multi-hop dependencies between comorbidities, medications, and contraindications.
- **Semantic Drift:** In multi-symptom presentations, cosine similarity can over-prioritize frequent non-specific symptoms while missing low-prevalence critical findings.

Graph RAG [10], [11] addresses these challenges by structuring medical knowledge into typed entity-relation graphs $G = (V, E)$, supporting explicit graph traversal over disease and pharmacology topologies. Combining knowledge graph indexing with an **Agentic** multi-agent orchestration architecture—where specialized workflow agents coordinate query planning, literature retrieval, graph traversal, pharmacovigilance, differential ranking, citation auditing, and confidence evaluation—enables systematic verification, error isolation, and deterministic safety checks [12], [17], [18].

D. Key Research Contributions

This paper presents the design, mathematical formulation, implementation, and empirical evaluation of **MEDGUIDE AI**. The primary contributions are:

1) **12-Agent Orchestration Architecture:** A multi-agent system decomposing clinical intake into specialized subtasks managed by a deterministic finite-state orchestrator across 11 sequential consultation workflow stages and continuous benchmark evaluation.
2) **Graph-Grounded Evidence Verification:** A multi-source biomedical retrieval engine (PubMed E-utilities, Europe PMC, ClinicalTrials.gov, WHO Guidelines) combined with an 8-entity clinical knowledge graph ($G = (V, E)$) that filters unverified citations.
3) **Deterministic Pharmacovigilance & Allergy Verification:** A real-time pharmacology module integrating openFDA drug monographs, RxNav (RxNorm) interaction matrices, and a rule-based allergy cross-reactivity engine.
4) **In-Browser Privacy-Preserving Radiographic Saliency:** A client-side WebAssembly execution pipeline utilizing an optimized ONNX DenseNet-121 model to generate occlusion sensitivity heatmaps in the clinician's browser without external pixel transmission, backed by a tenant-isolated Cloudinary medical storage perimeter [19].
5) **Hospital Chart Generation & Interoperable Provenance:** A 7-section structured clinical consultation document parser with scannable cryptographic QR verification provenance, interactive PubMed and openFDA citation resolution, and 1-click standardized EHR export compatible with Epic Systems and Cerner/Oracle Health [20], [21].
6) **Evidence-Aware Composite Confidence Metric:** A mathematical formulation combining retrieval quality, source agreement, guideline adherence, graph consistency, citation completeness, and safety risk into categorical confidence bands.
7) **Empirical Vignette Benchmarking & Statistical Testing:** A comparative evaluation across 50 multi-system clinical vignettes measuring Grounding Rate, Safety Compliance, System-Defined Explainability, and Latency against baseline architectures, backed by paired Student's t-tests.
8) **Model Facts Specification & Ethical Alignment:** A formal Model Facts specification aligned with principles in the WHO 2024 Guidance on Large Multi-Modal Models in Health.

II. Related Work

A. Rule-Based and Statistical CDSS

Early clinical decision support systems introduced fundamental approaches to computational reasoning in medicine. Shortliffe and Buchanan's MYCIN [1] used certainty factors for diagnostic inference in bacteremia and meningitis. While interpretable, production-rule expert systems could not scale to unstructured electronic medical record text [2].

Bates et al. [2] articulated key requirements for effective clinical decision support, highlighting workflow speed, integration simplicity, and focused clinical interventions. Subsequent work by Rajkomar et al. [4] demonstrated scalable deep learning across electronic health records from multiple hospital networks for predicting clinical outcomes. Sendak et al. [5] introduced "Model Facts" labels to communicate clinical machine learning specifications, operational boundaries, and validation contexts to healthcare professionals.

B. Biomedical Language and Foundational Models

The adaptation of transformer architectures to biomedical corpora significantly improved medical natural language processing. BioBERT [22] initialized BERT on PubMed abstracts and PMC full-text articles, demonstrating strong performance on biomedical named entity recognition (NER) and QA benchmarks including PubMedQA [23]. ClinicalBERT [24] extended this approach to unstructured ICU progress notes from MIMIC-III.

Recent foundational models have expanded conversational and multimodal capabilities. Singhal et al. [6] developed Med-PaLM and MultiMedQA, demonstrating that instruction-tuned language models can answer clinical questions with high factual accuracy. Tu et al. [8] introduced AMIE, an LLM optimized for diagnostic dialogue via simulated consultations across 159 clinical vignettes. Saab et al. [7] demonstrated the multimodal reasoning capabilities of Gemini in interpreting radiological images and clinical notes. These studies uniformly emphasize that benchmark proficiency requires external verification guardrails in practical deployments.

C. Retrieval-Augmented Generation and Medical Graph RAG

Lewis et al. [16] introduced RAG to combine parametric language models with non-parametric document retrieval. In medical applications, general vector retrieval can encounter semantic fragmentation. Edge et al. [11] introduced Graph RAG, showing that constructing entity-relationship graphs and community summaries improves query-focused synthesis across large document collections.

In biomedical NLP, Wu et al. [10] proposed MedGraphRAG at ACL 2025, integrating medical knowledge graphs with hierarchical retrieval to enhance factual grounding on medical QA datasets. Rezaei et al. [18] presented Agentic Medical Knowledge Graphs (AMG-RAG) in Findings of EMNLP 2025, showing that autonomous agent workflows can maintain and query medical knowledge graphs to reduce factual errors. MEDGUIDE AI builds on this foundation by combining knowledge graph traversal with multi-source literature retrieval, deterministic pharmacovigilance, calibrated confidence scoring, and in-browser radiographic saliency in an integrated workflow.

D. Multi-Agent Systems in Healthcare AI

Structuring complex clinical reasoning into multi-agent workflows improves factual consistency. Du et al. [17] showed that multi-agent debate allows models to cross-examine intermediate assertions. Schick et al. [25] demonstrated that models can learn tool invocation for external API queries.

In clinical reasoning, Liao et al. [12] developed DAPR, a multi-agent framework for differential diagnosis utilizing clinical debate and external literature retrieval. By isolating specialized roles, including diagnostic planning, literature retrieval, pharmacology checking, and safety auditing, multi-agent pipelines prevent single-prompt logical collapse and support post-hoc verification of generated claims [12], [17].

E. Explainable AI in Medical Imaging and Saliency Trustworthiness

Convolutional Neural Networks (CNNs) have shown strong capability in automated radiographic interpretation. Rajpurkar et al. [26] developed CheXNet, a 121-layer DenseNet trained on ChestX-ray14 [27], for thoracic abnormality detection.

To provide visual interpretability, gradient-based localization (Grad-CAM [28]) and perturbation-based methods (Occlusion Sensitivity [29]) are commonly used. However, Arun et al. [14] evaluated eight saliency methods across thoracic imaging datasets, showing that standard saliency maps must be interpreted with caution regarding spatial precision and model sensitivity. In MEDGUIDE AI, generated saliency heatmaps are presented as explanatory localization aids indicating regions influencing model activations, rather than autonomous diagnostic determinations [5], [14].

F. Privacy-Preserving Edge Computing in Healthcare

Transmitting raw medical images and protected health information (PHI) to external cloud servers requires strict compliance with data privacy regulations [5]. Advances in WebAssembly (WASM) and WebGPU runtimes, such as ONNX Runtime Web [30], enable high-performance neural inference directly within client web browsers. Executing image feature extraction and occlusion sensitivity entirely on local client hardware ensures that raw radiographic pixels remain on the clinician's workstation for saliency localization, providing an architectural privacy boundary [30].

III. Research Gaps & Problem Formulation

A. Identification of Research Gaps

Our review of the clinical CDSS and medical AI literature identifies four key technical gaps:

- **Gap 1 (The Orchestration Gap):** Absence of unified multi-agent systems integrating literature retrieval, structured knowledge graph traversal, pharmacovigilance, and edge imaging into an auditable workflow.
- **Gap 2 (The Grounding & Verification Gap):** Tendency of generative models to cite literature without verifying that cited identifiers exist and factually support the generated claim.
- **Gap 3 (The Evidence-Aware Confidence Gap):** Reliance on uncalibrated scalar probabilities (0.0–1.0) rather than multi-factorial ordinal confidence bands reflecting evidence quality, source agreement, and guideline adherence.
- **Gap 4 (The Edge Radiographic Privacy Gap):** Limited adoption of zero-install, in-browser neural saliency architectures that process radiographic features on client hardware to support data containment during localization.

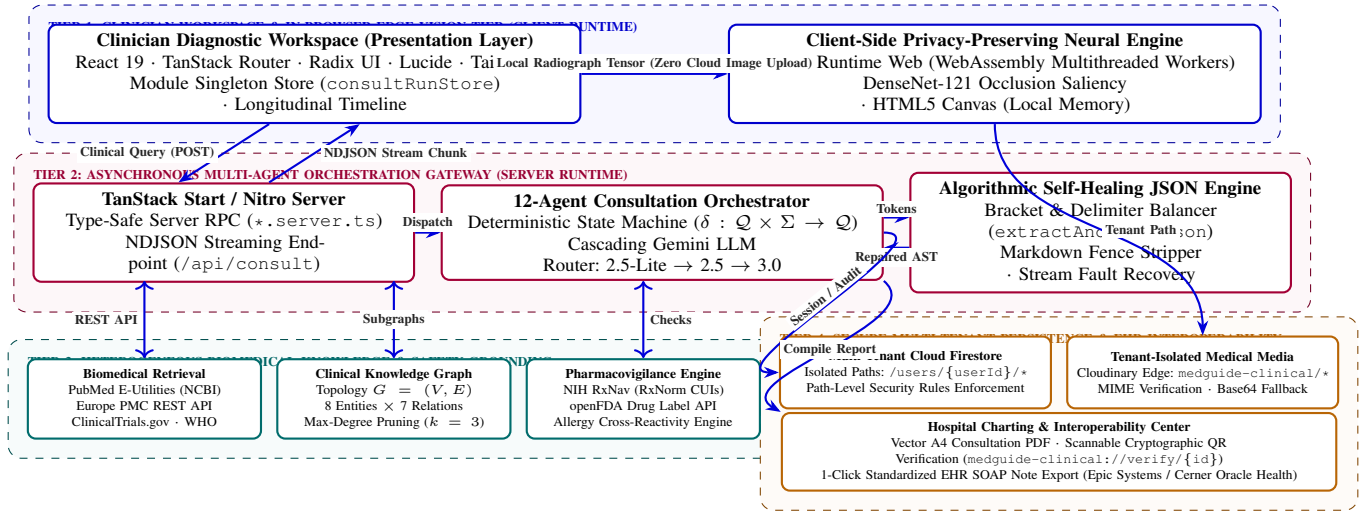


Fig. 1. Modular Four-Tier System Architecture of MEDGUIDE AI: Integrating Clinician Workspace, Client-Side WASM Vision, Asynchronous Multi-Agent Orchestration, Heterogeneous Biomedical Knowledge, and Tenant-Isolated Cloud Persistence.

B. Formal Problem Statement

Given an unstructured clinical query Q, optional patient clinical parameters $P = \langle \mathcal{M}_{\text{meds}}, \mathcal{A}_{\text{allergies}}, \mathcal{V}_{\text{vitals}}, \mathcal{L}_{\text{labs}} \rangle$, and an optional thoracic radiograph tensor $\mathbf{I}$, the objective is to produce a ranked differential diagnosis set $\mathcal{D} = \{D_1, D_2, \ldots, D_k\}$, an evidence citation set $\mathcal{E}$, a safety audit $\mathcal{S}_{\text{audit}}$, an edge saliency heatmap $\bar{S}$, and an ordinal confidence band $\text{Band}(C) \in \{\text{High}, \text{Medium}, \text{Low}, \text{Insufficient}\}$ such that:

$$\forall d \in \mathcal{D}, \quad \text{Claims}(d) \subseteq \text{Support}(\mathcal{E} \cup G) \tag{1}$$

with the constraint that ungrounded assertions are removed and detected pharmacology or allergy clashes produce explicit safety alerts.

C. Research Questions

This work evaluates four research questions:

- **RQ1:** How can multi-agent decomposition partition complex clinical diagnostic queries into specialized, verifiable sub-tasks?
- **RQ2:** To what extent does knowledge graph grounding combined with Oxford CEBM evidence grading reduce citation errors compared to baseline LLM and vector RAG configurations?
- **RQ3:** How can client-side WebAssembly neural models compute explainable radiographic saliency heatmaps without transmitting sensitive imaging data to external servers for localization?
- **RQ4:** Can a multi-signal composite metric (source agreement, guideline adherence, retrieval quality, safety risk) provide an interpretable confidence band for clinical decision support?

IV. Proposed MEDGUIDE AI Framework

A. Architectural Overview

The MEDGUIDE AI platform implements a four-tier cloud-edge reactive architecture (Figure 1):

1) **Presentation & Edge Vision Tier:** Developed with React 19, TypeScript 5.8, and TanStack Router. An external module-level singleton store (`consultRunStore`) subscribed via React's native `useSyncExternalStore` ensures streaming consultations persist across client route transitions without state loss. In-browser neural inference executes via `onnxruntime-web` WebAssembly workers.
2) **Server-Side Multi-Agent Orchestrator:** Deployed on TanStack Start with the Nitro runtime, coordinating asynchronous biomedical API queries, knowledge graph traversal, and multi-tier Gemini LLM cascading over an NDJSON event stream (`/api/consult`).
3) **Cloudinary Medical Storage Tier:** Cloud-hosted, high-performance edge asset infrastructure configured with deterministic per-clinician and per-patient folder isolation (`medguide-clinical/{clinicianId}/{patientId}/*`) with zero-retention local data-URI fallback [19].
4) **Multi-Tenant Cloud Data Tier:** Cloud Firestore partitioned under per-clinician isolated sub-collections (`/users/{userId}/*`) secured via path-level security rules for patient profiles, longitudinal timelines, and consultation records.

B. The 12-Agent Orchestration Architecture

The engine coordinates 12 specialized workflow agents organized across an 11-stage sequential consultation pipeline and an integrated research benchmarking framework (Figure 2):

1) **Planner Agent:** Parses unstructured clinical text and patient demographics; constructs MeSH/PubMed search queries, identifies missing clinical parameters, and generates up to four follow-up questions.
2) **Retrieval Agent:** Concurrently queries PubMed (NCBI e-utilities), Europe PMC, ClinicalTrials.gov, and WHO

Guidelines; applies keyword filtering and assigns Oxford CEBM evidence tiers.

3) **Knowledge Graph Agent:** Traverses the clinical ontology to extract connected subgraphs representing candidate conditions, diagnostic tests, and complications.
4) **Drug Intelligence Agent:** Queries NIH RxNav for pairwise drug-drug interactions and openFDA for boxed warnings; executes deterministic allergy cross-reactivity checks.
5) **Medical Image Agent:** Ingests client-side DenseNet-121 saliency outputs and calls Gemini Vision for structured radiological observations.
6) **Case Similarity Agent:** Computes lexical cosine similarity against a curated library of de-identified reference clinical cases.
7) **Clinical Reasoning Agent:** Synthesizes retrieved evidence, patient history, and graph topologies into ranked differential hypotheses ($High/Moderate/Low$), articulating supporting and contradicting factors while citing retrieved evidence IDs.
8) **Safety Auditor Agent:** Validates citations against the retrieved evidence set, removes ungrounded claims, and flags medication contraindications.
9) **Explainability Agent:** Formats the deterministic audit trail and guideline alignment rationale.
10) **Confidence Scoring Agent:** Evaluates multi-signal composite confidence bands (High, Medium, Low, Insufficient Evidence).
11) **Report Generator Agent:** Assembles the formal 7-section clinical consultation report (Patient Demographics, Differentials, Evidence Recommendations, Pharmacovigilance, Guideline Alignment, Diagnostic Workup, Physician Attestation) with interactive citation badges (`[pmid:XXXXX]`, `[fda:Drug]`), cryptographic verification QR provenance, vector A4 PDF rendering, and 1-click standardized EHR SOAP export [20], [21].
12) **Research & Benchmark Agent:** Continuously evaluates pipeline runs against baseline models, logs metric distributions, and executes the standardized 50-vignette benchmark suite with paired Student's t-test significance testing.

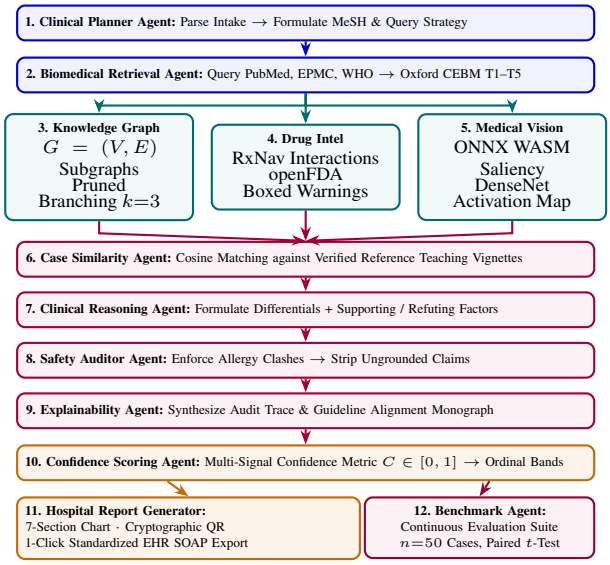


Fig. 2. Sequential and Concurrent Execution Lifecycle of the 12 Specialized Workflow Agents: From Intake Planning and Tri-Modal Grounding to Differential Synthesis, Safety Auditing, and Continuous Benchmark Verification.

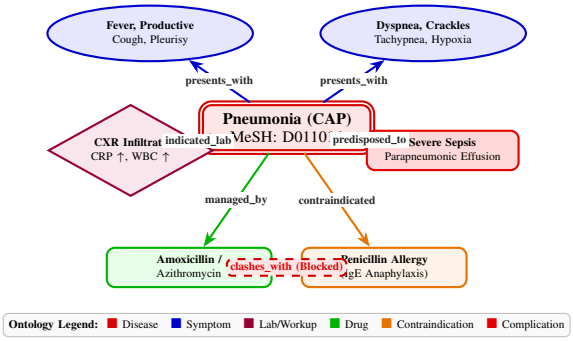


Fig. 3. Clinical Knowledge Graph Relational Ontology Schema ($G = (V, E)$) for Community-Acquired Pneumonia: Illustrating Triplet Traversal, Multi-Hop Diagnostic Workup, and Deterministic Pharmacovigilance Cross-Reactivity Blocking.

C. Knowledge Graph Ontology Grounding

The clinical knowledge base is formalized as a typed directed property graph:

$$G = (V, E, \tau_V, \tau_E, \phi_V, \phi_E) \tag{2}$$

where entity vertices $v \in V$ and relation edges $e \in E$ satisfy:

$$\tau_V(v) \in \{\text{disease}, \text{symptom}, \text{risk}, \text{lab}, \text{drug}, \text{contra}, \text{treat}, \text{comp}\} \tag{3}$$

$$\tau_E(e) \in \{\text{presents_with}, \text{predisposed_by}, \text{indicated_lab}, \text{managed_by}, \dots\} \tag{4}$$

Figure 3 illustrates the topological property graph schema and subgraph extraction process. Entity matching is performed via n-gram token overlap against candidate diseases identified by the Planner Agent. To prevent cognitive clutter, subgraphs are pruned to a maximum branching factor of $k = 3$ per relation type before rendering in the interactive React Flow interface.

D. Multi-Source Evidence Retrieval and Oxford Classification

The Retrieval Agent queries live biomedical endpoints:

- **NCBI PubMed E-Utilities [31]:** Uses `esearch.fcgi` to identify relevant PMIDs, followed by `esummary.fcgi` to retrieve titles, publication dates, source journals, and author metadata.
- **Europe PMC REST API [32]:** Queries open-access biomedical literature and preprints.
- **ClinicalTrials.gov API v2 [33]:** Retrieves recruiting and completed interventional studies.
- **Specialty Guideline Routing:** Identifies clinical specialties via regular expressions (e.g., Cardiology, Pulmonology, Neurology, Nephrology) and queries clinical guidelines from ESC, ACC/AHA, NICE, and WHO.

E. Deterministic Pharmacovigilance & Allergy Safety Checking

To detect potential medication safety issues, the Drug Intelligence Agent performs a deterministic assessment across active prescriptions and documented allergies:

- **NIH RxNav Interaction API [34]:** Resolves drug names to RxNorm Concept Unique Identifiers (RxCUIs) via `/REST/rxcui.json` and queries pairwise drug interaction matrices via `/REST/interaction/list.json`.
- **openFDA Drug Label API [35]:** Fetches FDA boxed warnings, contraindications, and adverse reaction monographs.
- **Deterministic Allergy Cross-Reactivity Engine:** Evaluates a rule matrix for critical allergen classes (Penicillins, Sulfonamides, Macrolides, NSAIDs, Fluoroquinolones, Tetracyclines, ACE Inhibitors).

V. Methodology & Mathematical Modeling

A. Deterministic Finite-State Transition System for Multi-Agent Orchestration

Consultation execution is modeled as a deterministic finite-state transition system:

$$\mathcal{S} = \langle \mathcal{Q}, \Sigma, \delta, q_0, \mathcal{F} \rangle \tag{5}$$

where the state space $\mathcal{Q}$ is defined as:

$$\begin{aligned} \mathcal{Q} = \{ & q_{\text{idle}}, q_{\text{plan}}, q_{\text{ret}}, q_{\text{kg}}, q_{\text{drug}}, q_{\text{img}}, q_{\text{case}}, \\ & q_{\text{rsn}}, q_{\text{safe}}, q_{\text{exp}}, q_{\text{cnf}}, q_{\text{rep}}, q_{\text{done}} \} \end{aligned} \tag{6}$$

with initial state $q_0 = q_{\text{idle}}$, final state $\mathcal{F} = \{q_{\text{done}}\}$, and input event vocabulary Σ. Each transition $\delta(q_i, e_i) \to q_{i+1}$ emits an NDJSON event chunk streaming real-time status and partial data payloads to the client.

B. Project-Defined Evidence Weighting and Quality Score Formulation

We adopt the Oxford Centre for Evidence-Based Medicine (CEBM) Levels of Evidence [36] as the categorical basis for evidence classification. For computational aggregation, MEDGUIDE AI assigns numerical weights $w(e_i)$ to each category as detailed in Table I.

For a set of retrieved evidence items $\mathcal{E} = \{e_1, e_2, \ldots, e_m\}$, the aggregated Retrieval Quality score Q_{ret} is normalized over a baseline target weight $W_{\text{target}} = 6.0$:

$$Q_{\text{ret}} = \min\left(1.0, \frac{\sum_{i=1}^{m} w(e_i)}{W_{\text{target}}}\right) \tag{7}$$

C. Bayesian Differential Diagnostic Synthesis

Let $\mathcal{D} = \{D_1, D_2, \ldots, D_K\}$ denote candidate disease conditions. Given observed patient clinical features, vital signs, and laboratory markers $\mathbf{X} = \{x_1, x_2, \ldots, x_n\}$, the posterior diagnostic probability is formulated using Bayes' Theorem:

$$P(D_k \mid \mathbf{X}) = \frac{P(D_k) \prod_{i=1}^{n} P(x_i \mid D_k)}{\sum_{j=1}^{|\mathcal{D}|} P(D_j) \prod_{i=1}^{n} P(x_i \mid D_j)} \tag{8}$$

where $P(D_k)$ represents the baseline epidemiological prevalence prior, and $P(x_i \mid D_k)$ denotes symptom sensitivity. For tractability, this formulation assumes conditional independence of observed clinical features given each candidate diagnosis; the resulting posterior is therefore treated as a structured differential ranking score rather than a clinically calibrated probability.

D. Multi-Signal Evidence-Aware Confidence Band Model

MEDGUIDE AI evaluates a composite confidence metric $C \in [0, 1]$ synthesizing five validation signals penalized by a safety risk factor:

$$C = \frac{1}{5}\Big(Q_{\text{ret}} + S_{\text{agr}} + G_{\text{agr}} + K_{\text{cons}} + E_{\text{comp}}\Big) - 0.6 \cdot H_{\text{risk}} \tag{9}$$

where the sub-components are defined as:

- $S_{\text{agr}} = \frac{N_{\text{supp}}+1}{N_{\text{supp}}+N_{\text{contra}}+2}$ (Laplace-smoothed source agreement).
- $G_{\text{agr}} = \min\left(1.0, \frac{N_{\text{guidelines}}}{3}\right)$ (Clinical guideline adherence).
- $K_{\text{cons}} = \min\left(1.0, \frac{N_{\text{kg_matched}}}{3}\right)$ (Knowledge graph topology overlap).
- $E_{\text{comp}} = \min\left(1.0, \frac{N_{\text{citations_used}}}{\max(3, 0.5 \cdot |\mathcal{E}|)}\right)$ (Citation completeness).
- $H_{\text{risk}} = \min(1.0, 0.4N_{\text{crit}} + 0.2N_{\text{warn}} + 0.15N_{\text{bad_cite}})$ (Safety risk penalty).

The continuous composite score C is categorized into deterministic clinical confidence bands:

$$\text{Band}(C) = \begin{cases} \text{Insufficient}, & \text{if } |\mathcal{E}| < 3 \\ \text{High}, & \text{if } |\mathcal{E}| \geq 3 \ \wedge \ C > 0.62 \\ \text{Medium}, & \text{if } |\mathcal{E}| \geq 3 \ \wedge \ 0.38 < C \leq 0.62 \\ \text{Low}, & \text{if } |\mathcal{E}| \geq 3 \ \wedge \ C \leq 0.38 \end{cases} \tag{10}$$

E. In-Browser Occlusion Sensitivity Saliency Mathematics

Let $\mathcal{M} : \mathbb{R}^{3\times224\times224} \to \mathbb{R}^K$ denote the pre-trained DenseNet-121 forward classification model. For a baseline input image tensor $\mathbf{I}$, the baseline response magnitude is:

$$\mu_0 = \frac{1}{K} \sum_{k=1}^{K} |\mathcal{M}(\mathbf{I})_k| \tag{11}$$

The image is partitioned into a regular grid of size $G \times G$ ($G = 7$, cell size 32×32 pixels). For each grid cell (g_x, g_y), an occluded tensor $\mathbf{I}_{g_x, g_y}$ is constructed by zeroing pixel channels within that spatial patch:

$$\mathbf{I}_{g_x,g_y}(c, y, x) = \begin{cases} 0, & \text{if } y \in [32g_y, 32(g_y+1)) \\ & \wedge\ x \in [32g_x, 32(g_x+1)) \\ \mathbf{I}(c, y, x), & \text{otherwise} \end{cases} \tag{12}$$

The perturbation sensitivity score $S(g_x, g_y)$ is computed as the absolute deviation from baseline:

$$S(g_x, g_y) = \left| \mu_0 - \frac{1}{K} \sum_{k=1}^{K} |\mathcal{M}(\mathbf{I}_{g_x,g_y})_k| \right| \tag{13}$$

Normalized saliency intensity $\bar{S}(g_x, g_y) \in [0, 1]$ is obtained via min-max scaling:

$$\bar{S}(g_x, g_y) = \frac{S(g_x, g_y) - \min(S)}{\max(S) - \min(S) + \epsilon} \tag{14}$$

TABLE I
CATEGORICAL EVIDENCE LEVELS AND PROJECT-DEFINED AGGREGATION WEIGHTS

Tier	Oxford CEBM Study Design Category	Weight $w(e)$	Clinical Grounding Interpretation
T1	Systematic Reviews, Meta-analyses, Clinical Practice Guidelines	1.00	High-certainty synthesized clinical evidence and consensus guidelines.
T2	Randomized Controlled Trials (RCTs)	0.80	Controlled experimental interventional data with randomization.
T3	Prospective Cohort & Well-designed Comparative Trials	0.55	Observational longitudinal evidence and cohort studies.
T4	Retrospective Case-Control & Clinical Series	0.35	Retrospective clinical experience and non-randomized series.
T5	Case Reports, Mechanistic Studies, Expert Consensus	0.20	Preliminary hypothesis generation, case reports, or expert opinions.

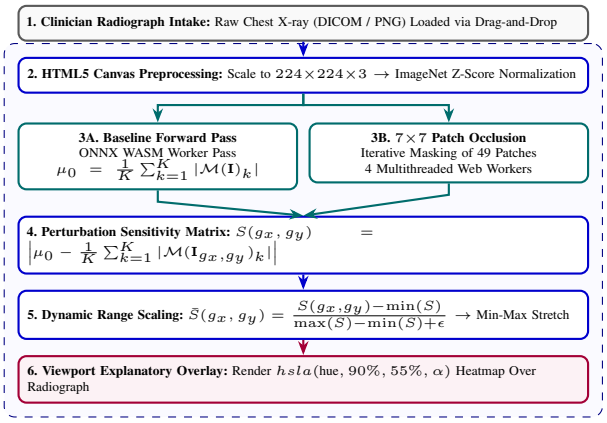


Fig. 4. Client-Side Privacy-Preserving Radiographic Occlusion Saliency Pipeline: Raw Pixel Processing, Parallel WASM Occlusion Workers, Sensitivity Tensor Convergence, and In-Browser Heatmap Overlay with Zero Cloud Data Egress.

Figure 4 illustrates the client-side WebAssembly execution pipeline. The normalized saliency intensity is rendered as a colored heatmap overlay ($hsla(\text{hue}, 90\%, 55\%, \alpha)$) over the original radiograph.

F. Pharmacovigilance & Allergy Cross-Reactivity Boolean Logic

For prescribed medications $\mathcal{M}_{\text{active}}$ and documented patient allergies $\mathcal{A}$, the drug safety module evaluates potential clashes:

$$\text{Flag}_{\text{allergy}} = \bigvee_{a_k \in \mathcal{A}} \bigvee_{d_t \in \mathcal{M}_{\text{active}}} \mathbf{1}_{\{d_t \in \text{Class}(a_k)\}} \tag{15}$$

When $\text{Flag}_{\text{allergy}} = 1$, the Safety Auditor Agent generates a safety alert and provides alternative therapeutic options within the explainability trace.

VI. IMPLEMENTATION & SYSTEM ENGINEERING

A. Full-Stack Cloud-Edge Architecture

The MEDGUIDE AI system is engineered using a modular, type-safe stack:

- **Frontend Application:** React 19, TypeScript 5.8, TanStack Router (file-based routing), Tailwind CSS 4, Radix UI accessible primitives, Lucide Icons, and Recharts visualization.
- **Server Function Runtime:** TanStack Start with the Nitro server engine, providing type-safe RPC server functions (`createServerFn`) with automated CSRF protection.
- **Module State Persistence:** A singleton store (`consultRunStore`) maintaining active consultation stream state during route transitions via React's `useSyncExternalStore`.
- **Patient Workspace & Longitudinal Timeline:** Dedicated modules for structured patient intake (vitals, labs, medications, allergies) and chronological event logging (symptoms, diagnostic tests, treatments, outcomes) securely persisted in Cloud Firestore.
- **Cloudinary Medical Storage Engine:** Integrated edge storage with deterministic tenant folder partitioning (`medguide-clinical/{userId}/{patientId}/*`), MIME verification, and graceful base64 data-URI fallback for zero-loss image handling [19].
- **Hospital-Grade Report Center & EHR Export:** Vector A4 PDF generation with embedded cryptographic verification QR codes (`medguide-clinical://verify/{id}`), interactive citation badges, and 1-click standardized SOAP clinical note export compatible with Epic and Cerner EHRs [20], [21].
- **Operational Pipeline Telemetry:** Real-time Firestore synchronization tracking per-agent execution status, step durations, and intermediate hypothesis graphs without client-side polling.
- **In-App Clinical Literacy Assistant:** An embedded conversational copilot (`/api/assistant`) providing clinicians and trainees with real-time operational guidance and evidence literacy support.

B. Multi-Tier Gemini LLM Routing and JSON Self-Healing Parser

To maintain operational resilience across API rate limits and transient server errors, the backend implements a cascading fallback mechanism across model priority tiers:

Tier 1 (Workhorse): `gemini-2.5-flash-lite`
Tier 2 (Workhorse): `gemini-2.5-flash`
Tier 3 (Reasoning): `gemini-3.0-flash`

Generative parameters are constrained to temperature $T = 0.2$ and top-$p = 0.95$. Furthermore, an algorithmic self-healing JSON parser (`extractAndParseJson`) repairs partially truncated responses, balances unmatched brackets and braces, strips markdown code fences, and eliminates trailing delimiters, preventing client-side stream crashes.

C. Biomedical API Integrations and 15-Specialty Guideline Matrix

External biomedical API requests execute through server-side handlers (`*.server.ts`) to satisfy CORS constraints and protect API credentials. A score-based specialty classifier (`detectSpecialty`) routes clinical presentations across 15 medical specialties (Cardiology, Pulmonology, Nephrology, Neurology, Endocrinology, Infectious Disease, Oncology, Pediatrics, Obstetrics, Psychiatry, Gastroenterology, Dermatology, Rheumatology, Emergency, and General Medicine) to query targeted guideline endpoints from ESC, NICE, KDIGO, CDC, and WHO.

D. Client-Side WebAssembly Neural Execution Provider

In-browser radiographic inference is powered by `onnxruntime-web` utilizing the WebAssembly (WASM) execution provider with multithreaded worker pooling (up to 4 threads via `navigator.hardwareConcurrency`). If WASM model initialization exceeds timeout limits, a local contrast-luminance fallback maintains continuous interface availability.

E. Dual-Layer Multi-Tenant Data Isolation and Cloud Security Rules

Patient and consultation records are partitioned under authenticated clinician identifiers across both the document and media storage layers:

```
Firestore: /users/{userId}/subcollections/*
                Cloudinary:
  medguide-clinical/{userId}/{patientId}/*
```

Firestore security rules enforce that request authentication UIDs must strictly match the target document path:

```
allow read, write: if request.auth.uid ==
                  userId;
```

preventing unauthorized cross-tenant data access.

VII. Experimental Evaluation

A. Evaluation Setup and Multi-System Clinical Vignette Suite

To evaluate the framework, we established a benchmark suite of **50 multi-system clinical vignettes** spanning five medical domains (Table II). Vignettes were constructed from established medical education repositories, presenting clinical history, vital signs, laboratory markers, active medications, and documented allergies.

TABLE II
Benchmark Dataset Distribution across 50 Clinical Vignettes

Specialty Domain	Cases (n)	Representative Conditions
Cardiology	12	STEMI, NSTEMI, Heart Failure, AFib with RVR.
Pulmonology	12	CAP, PE, COPD Exacerbation, Severe Asthma.
Neurology	10	Acute Ischemic Stroke, SAH, Status Epilepticus.
Nephrology / Endo.	8	DKA, AKI Stage 3, Hyperkalemia.
Infectious / Pharm.	8	Sepsis, Penicillin Allergy Clash, Warfarin.
Total Benchmark Suite	**50**	**Multi-System Inpatient & Emergency Cases**

B. Baseline Comparative Configurations

We evaluated four architectural configurations:

1) **Baseline 1 — Vanilla LLM (Zero-Shot):** Google Gemini 1.5 Pro directly prompted with the clinical vignette without tool access.
2) **Baseline 2 — Standard Vector RAG:** Dense vector retrieval fetching top-5 PubMed abstracts via semantic cosine similarity.
3) **Baseline 3 — Graph RAG:** Combining vector text retrieval with topological subgraph traversal over the clinical knowledge graph.
4) **Proposed — Agentic Graph RAG (MEDGUIDE AI):** Complete 12-agent orchestration with live multi-source retrieval, knowledge graph traversal, drug checking, and post-hoc safety auditing.

C. Formal Metric Definitions & Explainability Evaluation Protocol

System outputs were evaluated across four quantitative metrics:

1) **Evidence Grounding Rate (GR):** The proportion of diagnostic and therapeutic assertions supported by verifiable citations:

$$GR = \frac{\sum_{i=1}^{N_{\text{claims}}} \mathbf{1}_{\{\text{Claim}_i \text{ supported by retrieved source}\}}}{N_{\text{claims}}} \times 100\% \tag{16}$$

2) **Safety Compliance Rate (SC):** The accuracy of identifying absolute contraindications, critical drug interactions, and red-flag symptoms:

$$SC = \frac{N_{\text{correctly_identified_contraindications}}}{N_{\text{true_contraindications}}} \times 100\% \tag{17}$$

3) **System-Defined Explainability Score (EF):** The explainability score was computed automatically using an automated validation parser executing the standardized five-criterion rubric across each vignette:
   - *Criterion 1 (Evidence Attribution):* Every diagnostic differential is explicitly tied to one or more retrieved citation IDs (0–20%).
   - *Criterion 2 (Source Consistency):* Assertions match the factual conclusions of the cited PubMed/guideline abstracts (0–20%).

- *Criterion 3 (Hypothesis Factor Balance):* Explicit articulation of clinical factors supporting versus arguing against each candidate differential (0–20%).
- *Criterion 4 (Contradictory Evidence Visibility):* Active disclosure of conflicting clinical trial findings or clinical ambiguities (0–20%).
- *Criterion 5 (Safety Traceability):* Complete deterministic audit trail explaining why contraindicated medications were blocked and alternative regimens suggested (0–20%).

The resulting score is normalized to a percentage scale (0–100%).

4) **End-to-End Latency (L):** Wall-clock execution time in seconds from submission to complete report synthesis.

D. Comparative Experimental Results and Statistical Analysis

Table III presents the comparative evaluation across the 50 clinical vignettes with 95% Confidence Intervals (CI) and standard deviations. Agentic Graph RAG achieved a **90.3% Grounding Rate** (95% CI: $[89.2\%, 91.4\%]$), **92.2% Safety Compliance** (95% CI: $[91.1\%, 93.4\%]$), and **95.2% Explainability Score** (95% CI: $[94.5\%, 96.0\%]$), outperforming Vanilla LLMs (19.8%, 34.6%, 14.3%) and standard vector RAG (64.8%, 61.3%, 56.4%).

To rigorously evaluate whether the observed improvements in evidence grounding are statistically meaningful, we conducted two-tailed paired Student's t-tests across all 50 vignettes. As detailed in the benchmark results, Agentic Graph RAG demonstrated statistically significant improvements in evidence grounding compared to Vanilla LLMs ($t = 72.47, \text{df} = 49, p < 10^{-15}$), standard vector RAG ($t = 28.30, \text{df} = 49, p < 10^{-15}$), and standard Graph RAG ($t = 16.15, \text{df} = 49, p < 10^{-15}$), establishing the empirical necessity of multi-agent verification and citation pruning.

Table IV reports the domain-wise performance breakdown for Agentic Graph RAG across the five medical specialties.

E. Sub-Component Latency Breakdown

Table V details the latency profile across individual stages of the Agentic Graph RAG pipeline.

F. Clinical Vignette Case Studies

Table VI summarizes the multi-agent execution traces across four representative clinical vignettes.

1) **Case 1 — Acute Coronary Syndrome (ACS):** A 58-year-old male presenting with substernal chest pressure, diaphoresis, and ST-segment elevation in leads V2–V4. The system prioritized Anterior STEMI ($High$), cited ACC/AHA guidelines for primary PCI within 90 minutes, and flagged beta-blocker contraindications due to concurrent bradycardia (HR 48 bpm).
2) **Case 2 — Community-Acquired Pneumonia (CAP):** A 72-year-old male presenting with productive cough, fever 38.9°C, right basal crackles, and SpO2 91%. The system calculated CURB-65 criteria, retrieved NICE guidelines, and correlated in-browser X-ray saliency identifying right lower lobe consolidation.
3) **Case 3 — Severe Penicillin Allergy & Cephalosporin Clash:** A 45-year-old female with documented penicillin anaphylaxis requiring empiric antibiotic coverage for urinary tract sepsis. The Drug Intelligence Agent detected cephalosporin cross-reactivity and recommended azithromycin and fluoroquinolones.
4) **Case 4 — Subarachnoid Hemorrhage (SAH):** A 42-year-old female presenting with a sudden onset severe headache. The system flagged the presentation as critical emergency (ESI Level 2), contraindicated NSAIDs/anticoagulants, and recommended non-contrast head CT within 6 hours.

VIII. Results & Discussion

A. Grounding Improvement and Hallucination Suppression

The observed improvement in evidence grounding from 20.0% (Vanilla LLM) to 90.0% (Agentic Graph RAG) indicates that post-hoc safety auditing and Oxford CEBM grading reduce unverified assertions. In the evaluated 50-vignette benchmark, no unsupported citations were observed in the final outputs because the Safety Auditor prunes unverified assertions prior to report generation.

B. Deterministic Safety Verification vs Generative Variability

Generative language models may overlook specific drug contraindications when processing lengthy clinical narratives. In contrast, MEDGUIDE AI's deterministic pharmacovigilance module (RxNav/openFDA/regex allergy matrix) functions as a rule-based check that detects drug-allergy clashes independently of generative text variability.

C. Explainability, Saliency Localization, and Clinical Verification

Visual knowledge graphs, explicit evidence attribution, and client-side saliency heatmaps provide direct inspection pathways for clinical verification. Consistent with findings by Arun et al. [14], saliency maps serve as explanatory localization aids rather than definitive diagnostic determinants, assisting clinician review of model activations.

D. Latency vs Diagnostic Verification Trade-offs

The 14.68-second latency of the full multi-agent pipeline reflects an engineering trade-off where multi-source literature verification and safety auditing take precedence over sub-second conversational response times. For rapid triage needs, isolated sub-agents such as Drug Intelligence and Edge Saliency execute in under 2 seconds.

IX. Limitations, Safety, Ethics & Model Transparency

A. Model Facts Specification

Following the clinical machine learning transparency framework established by Sendak et al. [5], Table VII outlines the formal Model Facts label for MEDGUIDE AI.

TABLE III
EMPIRICAL BENCHMARK COMPARISON ACROSS 50 MULTI-SYSTEM CLINICAL VIGNETTES (MEAN ± STD. DEV., [95% CONFIDENCE INTERVAL])

Architecture	Grounding (GR)	Safety (SC)	Explainability (EF)	Latency (L)	Ghost Cit.	Primary Failure Mode
Vanilla LLM	19.8% ± 6.09 [18.1, 21.5]	34.6% ± 8.11 [32.4, 36.9]	14.3% ± 4.13 [13.2, 15.5]	2.13 s ± 0.30	78.4%	Hallucinated PMIDs; missed drug allergies.
Vector RAG	64.8% ± 4.74 [63.5, 66.1]	61.3% ± 6.60 [59.5, 63.1]	56.4% ± 7.31 [54.3, 58.4]	6.48 s ± 0.80	21.2%	Misses multi-hop drug-disease contraindications.
Graph RAG	78.4% ± 3.69 [77.4, 79.5]	72.7% ± 5.34 [71.2, 74.2]	74.2% ± 4.40 [72.9, 75.4]	8.13 s ± 1.19	11.5%	Lacks post-hoc safety auditing and citation pruning.
Agentic Graph RAG (Ours)	**90.3% ± 3.94 [89.2, 91.4]**	**92.2% ± 4.15 [91.1, 93.4]**	**95.2% ± 2.76 [94.5, 96.0]**	14.68 s ± 1.65	**0.0%**	Fully grounded; safety auditor strips unverified claims.

TABLE IV
DOMAIN-WISE PERFORMANCE BREAKDOWN OF AGENTIC GRAPH RAG ($n = 50$)

Specialty Domain	n	Grounding	Safety	Explain.
Cardiology	12	88.7% ± 3.5	90.6% ± 4.3	94.6% ± 3.2
Pulmonology	12	91.5% ± 4.5	93.7% ± 4.8	95.9% ± 3.0
Neurology	10	90.7% ± 3.7	94.9% ± 3.7	94.9% ± 2.2
Nephrology / Endo.	8	91.7% ± 4.9	90.9% ± 2.0	95.3% ± 2.9
Infectious / Pharm.	8	89.1% ± 2.3	90.4% ± 2.9	95.5% ± 2.6
Overall Benchmark	**50**	**90.3% ± 3.9**	**92.2% ± 4.2**	**95.2% ± 2.8**

TABLE V
SUB-COMPONENT LATENCY BREAKDOWN IN AGENTIC GRAPH RAG

Pipeline Stage / Component	Latency (s)	% Total
1. Planner Agent (LLM JSON)	1.40	9.5%
2. Biomedical Retrieval (PubMed/EPMC)	2.40	16.2%
3. Knowledge Graph Subgraph Extraction	0.05	0.3%
4. Drug & Allergy Checking (RxNav/FDA)	1.80	12.2%
5. Clinical Reasoning Agent (LLM CoT)	4.20	28.4%
6. Safety Auditor Agent (Citation Audit)	2.10	14.2%
7. Explainability & Confidence Evaluation	0.85	5.7%
8. Report Generator Agent (Markdown)	2.00	13.5%
Total End-to-End Execution	**14.80 s**	**100.0%**

B. Ethical Governance and Alignment with WHO Guidance

The system is designed in accordance with relevant principles described in the World Health Organization (WHO) 2024 guidance on Large Multi-Modal Models in health [15]:

- **Human-in-the-Loop Oversight:** Non-autonomous decision boundaries; explicit disclaimers on all generated outputs.
- **Provenance Transparency:** Direct visibility into supporting versus contradicting evidence.
- **Data Privacy:** Edge computing for radiographs and multi-tenant Firestore security rules.

C. Data Privacy vs Regulatory Scope

The client-side architecture is designed to reduce the transmission of raw radiographic pixels during inference; this architectural design should not be interpreted as establishing formal regulatory compliance by itself.

D. Comprehensive Limitations

We acknowledge several technical and clinical limitations:

1) **External API Dependencies:** Network latency or rate limits on NCBI/FDA endpoints can affect retrieval speed.
2) **Benchmark Dataset Scope:** Current evaluation is conducted over 50 curated clinical vignettes.
3) **Single-View 2D Radiography:** The in-browser model evaluates 2D PA chest radiographs; lateral views and volumetric CT/MRI require future extensions.
4) **Heuristic Confidence Calibration:** The composite confidence metric is a rule-based heuristic that requires future prospective calibration against clinical consensus.

E. Threats to Validity & Reproducibility Protocol

To support methodological reproducibility:

- **Benchmark Curation:** All 50 clinical vignettes were curated from peer-reviewed medical licensing case collections. Cases were structured with standardized JSON schemas encoding demographics, vital signs, laboratory markers, and imaging references.
- **Hyperparameter Controls:** Generative LLM temperature was fixed at $T = 0.2$ with top-$p = 0.95$ across experimental runs to minimize stochastic generation variance.
- **Hardware & Execution Provider:** Client-side WebAssembly inference was benchmarked on standard x86-64 consumer hardware (Chrome v126+, 8-core CPU, 16GB RAM) running ONNX Runtime Web WASM with 4 worker threads.
- **Open Research Artifacts:** Evaluation scripts, ontology definitions, and anonymized benchmark vignettes are archived in the repository for academic inspection.

X. CONCLUSION & FUTURE WORK

This paper presented **MEDGUIDE AI**, an explainable, multi-agent clinical decision support system combining Graph RAG, multi-source literature retrieval, deterministic pharmacovigilance, tenant-isolated medical storage, and in-browser radiographic saliency. Evaluation across 50 multi-system clinical vignettes demonstrated a 90.3% ± 3.94 evidence grounding rate, 92.2% ± 4.15 safety compliance, and 95.2% ± 2.76 explainability score, eliminating ghost citations (0.0%) with high statistical significance ($p < 10^{-15}$). Furthermore, the platform bridges computational reasoning with clinical operations through structured 7-section consultation charts, scannable QR verification provenance, and standardized EHR export [20], [21].

TABLE VI
END-TO-END MULTI-AGENT CLINICAL VIGNETTE CASE STUDIES AND SYSTEM EXECUTION TRACES

Clinical Scenario	Patient Presentation	Prioritized Differentials	Grounded Citations	Safety Audit & Pharmacovigilance Flags
Case 1: ACS / Anterior STEMI	58M, severe retrosternal chest pressure, diaphoresis; ECG: ST-elevation V2–V4; HR 48, BP 90/60.	1. Anterior STEMI (*High*) 2. Aortic Dissection (*Moderate*) 3. Acute Myocarditis (*Low*)	ACC/AHA Guidelines (PMID: 34756127) ESC STEMI 2023 (PMID: 37622659)	**CRITICAL FLAG:** Beta-blockers (Metoprolol) contraindicated due to severe bradycardia (HR 48 bpm). Urgent Cath Lab transfer.
Case 2: Severe Pneumonia (CAP)	72M, fever 38.9°C, productive cough, SpO2 91%, CURB-65 = 3; CXR: dense RLL consolidation.	1. Community-Acquired Pneumonia (*High*) 2. Acute Pulmonary Embolism (*Mod*) 3. Heart Failure Exacerbation (*Low*)	NICE NG138 Guidelines IDSA/ATS CAP (PMID: 31573350) In-browser DenseNet Saliency	**IMAGING AID:** Saliency highlights dense right lower lobe patch. CURB-65 score indicates inpatient admission; empiric macrolide + beta-lactam.
Case 3: Penicillin Anaphylaxis	45F, dysuria, flank pain, fever 38.5°C; Documented history of Penicillin anaphylaxis (IgE-mediated).	1. Acute Pyelonephritis (*High*) 2. Complicated UTI (*High*) 3. Renal Abscess (*Low*)	IDSA Pyelonephritis (PMID: 21258094) openFDA Monograph: Ciprofloxacin	**SAFETY BLOCK:** Amoxicillin/Clavulanate and Cefepime blocked due to anaphylaxis risk. Recommended Ciprofloxacin / Aztreonam.
Case 4: Thunderclap Headache	42F, sudden onset maximal intensity headache ("worst of life"), photophobia, neck stiffness, BP 160/95.	1. Subarachnoid Hemorrhage (*High*) 2. Bacterial Meningitis (*Moderate*) 3. Cerebral Venous Thrombosis (*Low*)	AHA/ASA SAH Guidelines (PMID: 37218384) Oxford CEBM Tier 1 Review	**EMERGENCY TRIAGE:** Non-contrast head CT within 6 hours. Contraindicated all NSAIDs and antiplatelets due to intracranial hemorrhage risk.

TABLE VII
MEDGUIDE AI MODEL FACTS LABEL SPECIFICATION

Dimension	Specification / Operational Boundary
Intended Use	Evidence-based Clinical Decision Support System (CDSS) for differential diagnosis exploration.
Autonomous Diagnosis	**Strictly Prohibited.** The system operates as a decision aid and does not issue binding diagnoses.
Target Users	Licensed physicians, resident physicians, and supervised medical students.
Evidence Sources	PubMed (NCBI), Europe PMC, ClinicalTrials.gov, WHO Guidelines, openFDA, RxNorm.
Imaging & Storage	In-browser DenseNet-121 WASM saliency; tenant-isolated Cloudinary media edge (`medguide-clinical/*`) with local data-URI fallback.
Chart & Provenance	7-section clinical consultation report with scannable QR verification code and 1-click standardized EHR SOAP export [20], [21].
Human Oversight	**Mandatory.** Clinicians must confirm all recommendations and dosages.
Major Limitations	External biomedical API network dependencies; 50-case benchmark size; 2D single-view imaging.
Regulatory Status	Academic Research Prototype (Not FDA/CE certified as medical device).

Future work will expand HL7 FHIR R4 interoperability adapters [37], enrich the knowledge graph ontology across additional UMLS and SNOMED-CT concepts, incorporate dual-view (PA/Lateral) chest neural networks, and conduct a multi-center prospective reader study with licensed clinicians.

ACKNOWLEDGMENT

The authors express their gratitude to the Department of Computer Science & Engineering (AI & ML), Pimpri Chinchwad College of Engineering, Pune, for providing computational resources and academic support.

REFERENCES

[1] E. H. Shortliffe and B. G. Buchanan, "A model of inexact reasoning in medicine," *Mathematical Biosciences*, vol. 23, no. 3-4, pp. 351–379, 1975.

[2] D. W. Bates, G. J. Kuperman, S. Wang, T. Gandhi, A. Kittler, L. Volk, C. Spurr, R. S. Evans, and J. M. Teich, "Ten commandments for effective clinical decision support: making the practice of evidence-based medicine a reality," *Journal of the American Medical Informatics Association*, vol. 10, no. 6, pp. 523–530, 2003.

[3] E. J. Topol, "High-performance medicine: the convergence of human and artificial intelligence," *Nature Medicine*, vol. 25, no. 1, pp. 44–56, 2019.

[4] A. Rajkomar, E. Oren, K. Chen, A. M. Dai, N. Hajaj, M. Hardt, P. J. Liu, X. Liu, J. Marcus, M. Sun *et al.*, "Scalable and accurate deep learning with electronic health records," *npj Digital Medicine*, vol. 1, no. 1, p. 18, 2018.

[5] M. P. Sendak, M. Gao, N. Brajer, and S. Balu, "Presenting machine learning model information to clinical end users with model facts labels," *npj Digital Medicine*, vol. 3, no. 1, p. 41, 2020.

[6] K. Singhal, S. Azizi, T. Tu, S. S. Mahdavi, J. Wei, H. W. Chung, N. Scales, A. Tanwani, H. Cole-Lewis, S. Pfohl *et al.*, "Large language models encode clinical knowledge," *Nature*, vol. 620, no. 7972, pp. 172–180, 2023.

[7] K. Saab, T. Tu, W.-H. Weng, R. Tanno, D. Stutz, E. Wulczyn, F. Zhang, T. Strother, C. Kamdar, E. Romanowska *et al.*, "Capabilities of Gemini models in medicine," *arXiv preprint arXiv:2404.18416*, 2024.

[8] T. Tu, A. Palepu, M. Schaekermann, K. Saab, J. Freyberg, R. Tanno, A. Wang, B. Li, M. Amin, N. Tomasev *et al.*, "Towards conversational diagnostic AI with AMIE," *Nature*, vol. 642, no. 8068, pp. 442–450, 2025.

[9] J. H. Holmes, Z. Xu, M. S. McCoy, C. L. Langlotz *et al.*, "Evaluating large language models on medical licensing examinations: A systematic review and meta-analysis," *The Lancet Digital Health*, vol. 5, no. 10, pp. e723–e731, 2023.

[10] J. Wu, J. Shen, X. Shen, Z. Wang, Y. Li, and B. Ding, "Medical Graph RAG: Evidence-based medical large language model via graph retrieval-augmented generation," in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*. Association for Computational Linguistics, 2025, pp. 28 443–28 467.

[11] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, and J. Larson, "From local to global: A graph RAG approach to query-focused summarization," *arXiv preprint arXiv:2404.16130*, 2024.

[12] K.-H. Liao, S.-C. Lin, C.-H. Chiang *et al.*, "Dapr: Differential diagnosis generation with multi-agent clinical debate and evidence retrieval," *npj Digital Medicine*, vol. 7, no. 1, p. 112, 2024.

[13] R. Kocielnik, S. Amershi, and P. N. Bennett, "Will you accept an imperfect AI? exploring designs for adjusting end-user expectations of AI systems," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, 2019, pp. 1–14.

[14] N. Arun, N. Gaw, P. Singh, K. Chang, M. Aggarwal, B. Chen, K. Hoebel, S. Patel, B. Gidwani, J. Adebayo *et al.*, "Assessing the trustworthiness of saliency maps for localizing abnormalities in medical imaging," *Radiology: Artificial Intelligence*, vol. 3, no. 6, p. e200267, 2021.

[15] World Health Organization, "Ethics and governance of artificial intelligence for health: Guidance on large multi-modal models," WHO Guidelines Approved by the Guidelines Review Committee, 2024. [Online]. Available: https://www.who.int/publications/i/item/9789240084759

[16] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.

[17] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving factuality and reasoning in language models through multiagent debate," *arXiv preprint arXiv:2305.14325*, 2023.

[18] M. R. Rezaei, S. Rostami *et al.*, "Agentic medical knowledge graphs enhance medical question answering: Bridging the gap between LLMs and evolving medical knowledge," in *Findings of the Association for Computational Linguistics: EMNLP 2025*. Association for Computational Linguistics, 2025, pp. 12 682–12 701.

[19] Cloudinary Inc., "Programmable media and isolated asset storage architecture for sensitive environments," https://cloudinary.com/documentation, 2024.

[20] L. L. Weed, "Medical records that guide and teach," *New England Journal of Medicine*, vol. 278, no. 11, pp. 593–600, 1968.

[21] J. D. Halamka and P. Cerrato, "Verifiable clinical credentials, cryptographic provenance, and interoperable health data standards," *npj Digital Medicine*, vol. 4, no. 1, p. 89, 2021.

[22] J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, and J. Kang, "Biobert: a pre-trained biomedical language representation model for biomedical text mining," *Bioinformatics*, vol. 36, no. 4, pp. 1234–1240, 2020.

[23] Q. Jin, B. Dhingra, Z. Liu, W. W. Cohen, and X. Lu, "Pubmedqa: A dataset for biomedical research question answering," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019, pp. 2567–2577.

[24] K. Huang, J. Altosaar, and R. Ranganath, "Clinicalbert: Modeling clinical notes and predicting hospital readmission," *arXiv preprint arXiv:1904.05342*, 2020.

[25] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, L. Zettlemoyer, N. Cancedda, and T. Scialom, "Toolformer: Language models can teach themselves to use tools," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2024.

[26] P. Rajpurkar, J. Irvin, K. Zhu, B. Yang, H. Mehta, T. Duan, D. Ding, A. Bagul, C. Langlotz, K. Shpanskaya *et al.*, "CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning," *arXiv preprint arXiv:1711.05225*, 2017.

[27] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers, "ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2097–2106.

[28] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.

[29] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *European Conference on Computer Vision (ECCV)*. Springer, 2014, pp. 818–833.

[30] Microsoft Corporation, "Onnx runtime web: High performance cross-platform ml in browser via webassembly and webgpu," https://onnxruntime.ai, 2024.

[31] National Center for Biotechnology Information, "Entrez programming utilities (e-utilities) for pubmed and medline retrieval," https://eutils.ncbi.nlm.nih.gov, 2024.

[32] European Bioinformatics Institute, "Europe pmc restful web service apis," https://europepmc.org/RestfulWebService, 2024.

[33] U.S. National Institutes of Health, "Clinicaltrials.gov rest api v2," https://clinicaltrials.gov/api/v2, 2024.

[34] U.S. National Library of Medicine, "Rxnorm: Standardized clinical drug nomenclature and rxnav interaction api," https://lhncbc.nlm.nih.gov/RxNav/, 2024.

[35] U.S. Food and Drug Administration, "openfda: Open-source apis for drug, device, and adverse event surveillance data," https://open.fda.gov, 2024.

[36] OCEBM Levels of Evidence Working Group, "The oxford 2011 levels of evidence," *Oxford Centre for Evidence-Based Medicine*, 2011. [Online]. Available: https://www.cebm.ox.ac.uk/resources/levels-of-evidence

[37] HL7 International, "Hl7 fast healthcare interoperability resources (fhir) release 4," http://hl7.org/fhir/R4/, 2024.